# $3d^1$

June 11, 2026

## 1 $\left|3d_{xy}^{\alpha}3d_{xz}^{0}3d_{yz}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$

Here are the CSFs:

$$|1,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$
$$|2,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$
$$|3,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$
$$|4,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$
$$|5,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|6,1/2,-1/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$
$$|7,1/2,-1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$
$$|8,1/2,-1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$
$$|9,1/2,-1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$
$$|10,1/2,-1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

The Ground State CSFs are 3 8

$\left\langle 1,\frac{1}{2},\frac{1}{2}\left|L_x\right|3,\frac{1}{2},\frac{1}{2}\right\rangle = I$

$\Delta g_{xx}$

$-2\zeta\Delta_{3-1}^{-1}$

$\left\langle 2, \frac{1}{2}, \frac{1}{2} \left| L_y \right| 3, \frac{1}{2}, \frac{1}{2} \right\rangle = -I$

$\Delta g_{yy}$

$-2\zeta\Delta_{3-2}^{-1}$

$\left\langle 5, \frac{1}{2}, \frac{1}{2} \left| L_z \right| 3, \frac{1}{2}, \frac{1}{2} \right\rangle = -2I$

$\Delta g_{zz}$

$-8\zeta\Delta_{3-5}^{-1}$